

RUIQI (RICKEY) HUANG

Atlanta, GA | ✉ rickey.huang747@gmail.com | ☎ (336) 770-0102

🌐 sites.gatech.edu/rickeyhuang77 | 🔗 linkedin.com/in/ruiqi-huang-44968a1a2 | 🐙 github.com/Rick3yHuang

PROFILE

Mathematics Ph.D. candidate with expertise in computational algebraic geometry, algorithm development, and scientific computing. Uses Python, Macaulay2, and MATLAB to translate mathematical models into reproducible computational methods for complex research and engineering problems.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Mathematics

August 2022 – Present

Wake Forest University

Winston-Salem, NC

B.S. in Mathematics; Minors in Statistics and Computer Science

August 2019 – May 2022

Honors in Mathematics; Dean's List (2019–2022)

RESEARCH EXPERIENCE

Computing and Stabilizing Ideals of Jukes-Cantor Network Models

July 2024–August 2026

- Developed and implemented an algorithm that automatically parameterizes Jukes–Cantor phylogenetic networks of arbitrary size and level in Fourier coordinates, replacing manual construction with symbolic adjacency-matrix path encoding and reducing per-tree computation from $\mathbf{O(n^2)}$ to $\mathbf{O(n)}$.
- Applied the parameterization algorithm across Jukes–Cantor network families, using computational patterns to formulate conjectures and establish stabilization theorems for dimensions and ideals.

Approximating Periodic Orbits and Navigation Minimal Problems

September 2025–May 2026

- Built compact quartic and sextic implicit models for Earth–Moon Lyapunov and Halo orbits, addressing the lack of closed-form orbit formulas in circular restricted three-body dynamics by fitting JPL catalogs via least squares and using the Jacobi constant—an energy-like invariant—to parameterize the orbit families.
- Developed a Macaulay2 toolkit to construct CR3BP liaison-navigation minimal problems—the smallest measurement sets yielding finitely many candidate spacecraft states—from algebraic orbit models and range, range-rate, or line-of-sight data, and compute their algebraic degrees.

Projective Plane Subdivision for Initial Orbit Determination

August 2022–September 2025

- Developed a candidate-evaluation routine for time-free, angles-only orbit determination from five 3D lines of sight, using two polynomial consistency checks to test whether a candidate orbital plane yields an orbit with the central body at a focus.
- Designed adaptive triangulation of the real projective plane using the evaluator's residual outputs and analytic Jacobian in physics-based, Newton-based, and gradient-based oracles, narrowing candidate regions to **0.10%–2.09% of the search domain** across three experiments.
- Built certified existence and exclusion oracles using interval arithmetic and the Krawczyk operator to accept projective-plane regions proven to contain at least one solution and reject regions proven solution-free.

INTERNSHIP EXPERIENCE

Beijing ByteDance Technology Co., Ltd.

Beijing, China

Educational Researcher, Gauthmath

December 2020 – July 2021

- Designed 50+ algorithmic solution workflows for a math education product, Gauthmath, translating math reasoning into implementation-ready specifications and partnered with engineers to implement them.
- Curated and annotated 2,500+ calculus problems for model training, standardizing step-by-step solutions to cover **10.43% more** user-submitted questions with readable solution paths.
- Analyzed 300+ negative user reports to isolate recurring failure modes and translate user feedback into targeted algorithm-improvement proposals.

TECHNICAL SKILLS

Programming and computation: Python, R, Java, MATLAB, Macaulay2

Tools: $\text{\LaTeX}$, R Markdown, Git

Languages: Mandarin (native), English (professional)